

→ VIEW FILE IDEA:

① UPLOAD PAGE

REF HUNTERS	
+ NEW CHAT	
	WELCOME!
	CHOOSE THE EXPERTISE LEVEL BEGINNER INTERMEDIATE ADVANCED
	ASK ABOUT...  
	ENTER FILE or DOI
	QUESTION'S EXAMPLE SUMMARIZE THE PAPER LIST A CITED PAPER

② ANSWER and VIEW FILE PAGE

REF HUNTERS	NAME of THE FILE	
+ NEW CHAT	< 2 of 25 >	CHAT BOT YOUR QUESTION
	RESEARCH ARTICLE Pluripotency, Differentiation, and Reprogramming: A Gene Expression Dynamics Model with Epigenetic Feedback Regulation Tadashi Miyamoto¹, Chikara Furusawa², Kunihiko Kaneko^{1*} ¹ Department of Basic Science, The University of Tokyo, Meguro-ku, Tokyo, Japan, ² Quantitative Biology Center (QBC), RIKEN, Saita, Osaka, Japan * kaneko@complex.c.u-tokyo.ac.jp Abstract Embryonic stem cells exhibit pluripotency: they can differentiate into all types of somatic cells. Pluripotent genes such as Oct4 and Nanog are activated in the pluripotent state, and their expression decreases during cell differentiation. Inversely, expression of differentiation genes such as Gata6 and Gata4 is promoted during differentiation. The gene regulatory network controlling the expression of these genes has been described, and slower-scale epigenetic modifications have been uncovered. Although the differentiation of pluripotent stem cells is normally irreversible, reprogramming of cells can be experimentally manipulated to regain pluripotency via overexpression of certain genes. Despite these experimental advances, the dynamics and mechanisms of differentiation and reprogramming are not yet fully understood. Based on recent experimental findings, we constructed a simple gene regulatory network including pluripotent and differentiation genes, and we demonstrated the existence of pluripotent and differentiated states from the resultant dynamical-systems model. Two differentiation mechanisms, interaction-induced switching from an expression	CAN YOU SUMMARIZE THE FILE AND ADD INFO FROM INTERNET? The paper by Miyamoto, Furusawa, and Kaneko (2015) presents a dynamical-systems model that links gene-expression oscillations, cell-cell interactions, and epigenetic feedback to explain how embryonic stem cells maintain pluripotency, differentiate, and can be reprogrammed. In the model, pluripotent genes (e.g., Oct4, Nanog) exhibit sustained oscillations in a single-cell "stem" state. As cell numbers increase, diffusive coupling of the differentiation gene product (modeled as gene x4) desynchronizes these oscillations, causing some cells to switch to a fixed-point expression pattern that suppresses pluripotent genes and activates differentiation genes (Gata6, Gata4). This "interaction-induced switching" is robust to moderate Gaussian noise ($\sigma \leq 0.1$). Epigenetic variables $\theta_{ik}(t)$ act as slowly evolving thresholds ($\tau_{ep} \approx 10^3$) that lock the chosen state; once fixed, the differentiated state resists perturbations and forms an epigenetic barrier to pluripotency. Reprogramming is achieved by prolonged overexpression of multiple pluripotent factors—corresponding to the Yamanaka factors Oct4, Sox2, Klf4, and Myc—long enough to reduce the epigenetic thresholds, allowing oscillations (and thus pluripotency) to re-emerge. The model therefore predicts that both the creation of oscillatory gene expression and the relaxation of epigenetic fixation are required for successful induction of iPSC cells. ASK THIS FILE A QUESTION...  
		HERE IS EXACTLY THE SNIPPET of THE FILE UPLOADED PAGE: 1 PARAGRAPH: 1 LINE: 6-12 CITATION 1: PAGE: 1 LINE: 3-6 CITATION 2: ... + HIGHLIGHTED PART IN THE PAPER HERE PUT EXACTLY THE SNIPPET of ARTICLE FROM ONLINE SOURCE → THE NAME of PAPER SECTION → INTRODUCTION VIEW ONLINE ↓ MAYBE COULD SHOW WE CANNOT OPEN IT ③ EXPLORATION PAGE